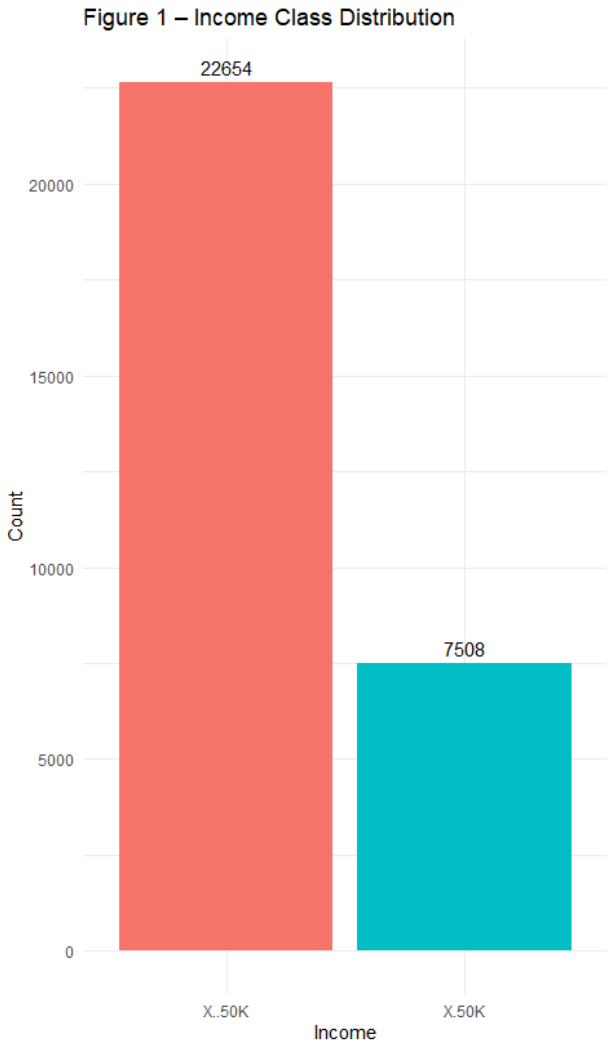
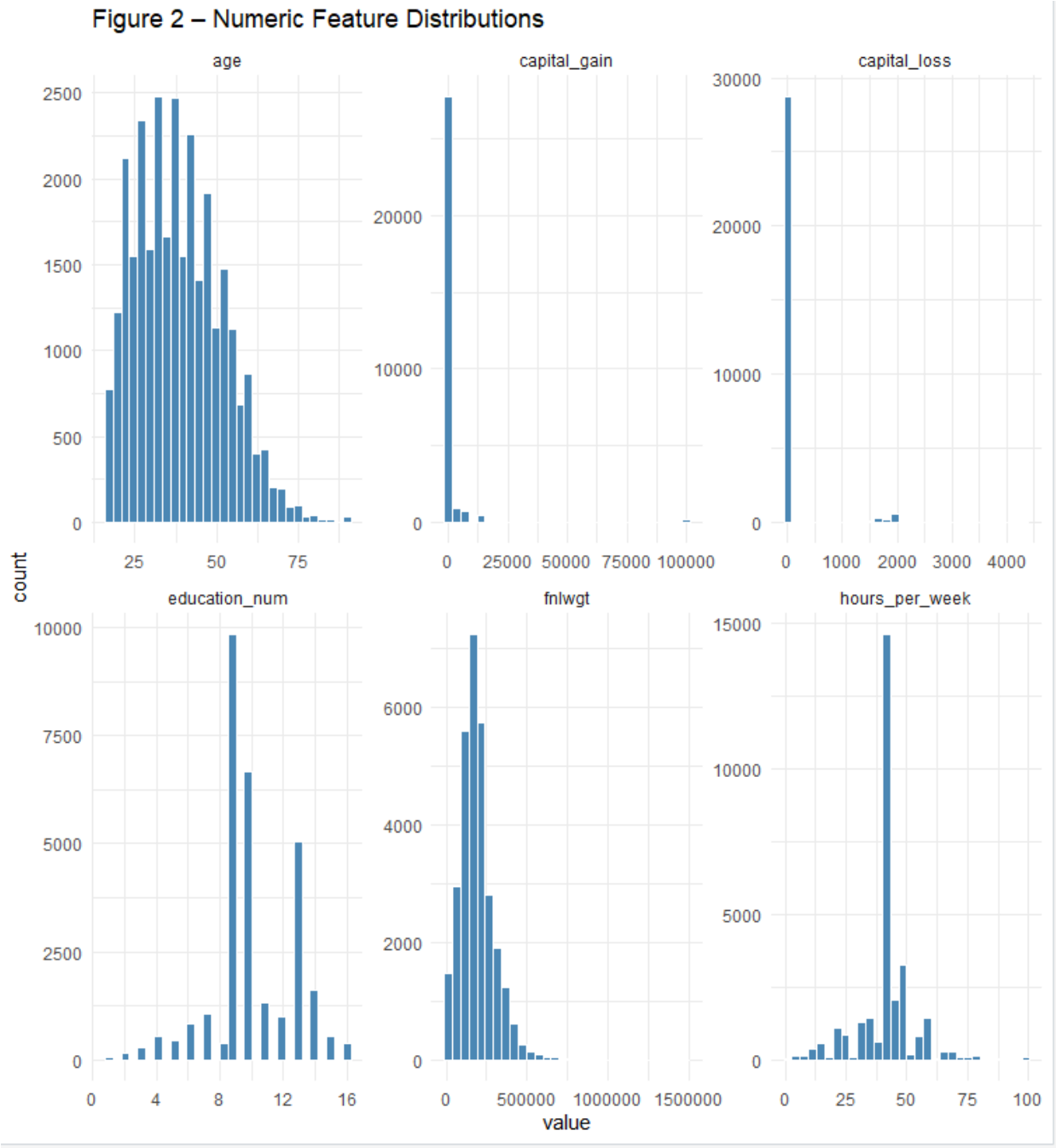
1. Exploratory Data Analysis

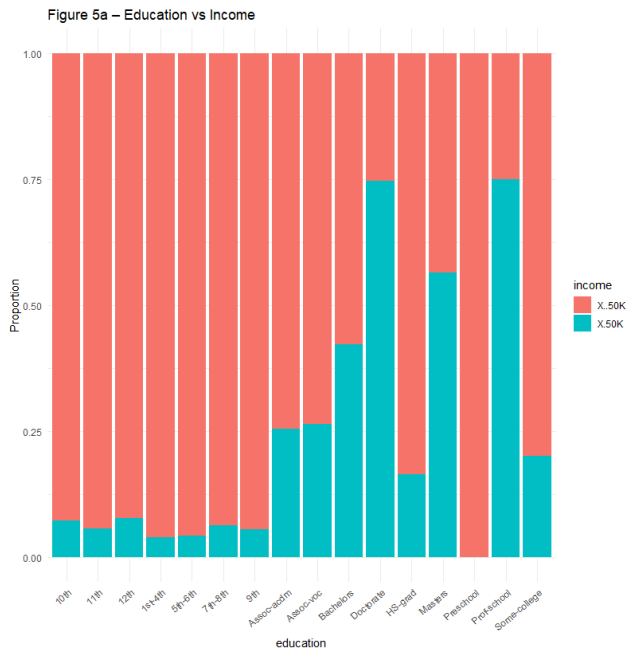
The UCI Adult Census Income dataset contains 48,842 records of demographic and employment information from the 1994 US Census. Each record has 14 features including age, education, occupation, and capital gains, with the goal of predicting whether an individual earns above or below $50,000 per year. After removing rows with missing values (encoded as "?"), 45,222 complete records remained.

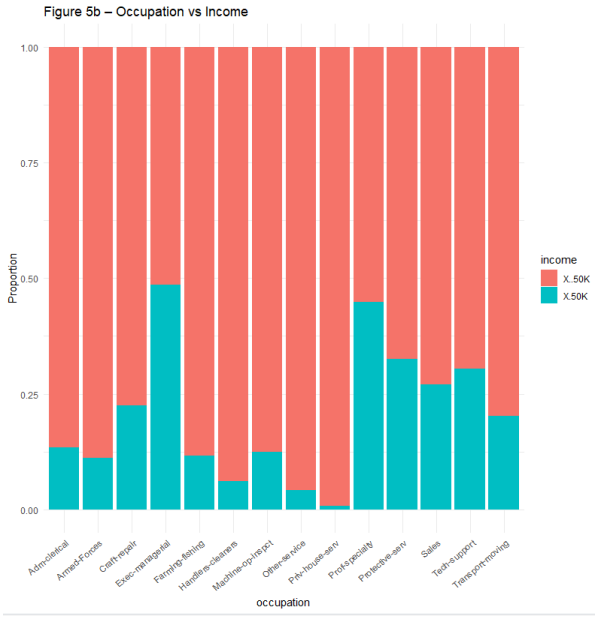


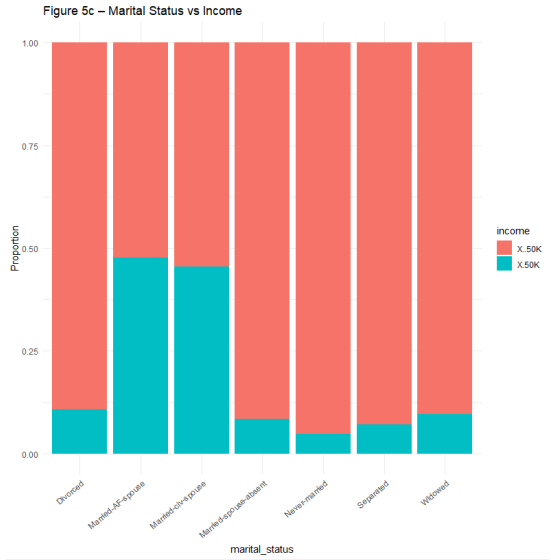
The target variable is imbalanced, with 76% of individuals earning under $50K and 24% earning above. This means a naive classifier predicting the majority class would still achieve 76% accuracy, so accuracy alone is an insufficient evaluation metric.

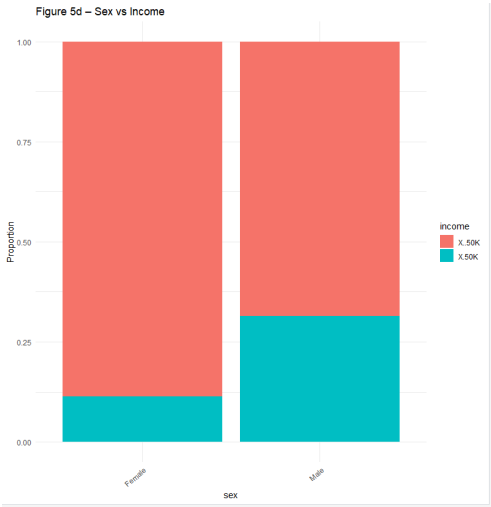


Age and hours-per-week are roughly normally distributed. Capital-gain and capital-loss are heavily right-skewed, with most values at zero and a small number of extreme outliers. These two variables were Winsorised at the 1st and 99th percentiles to reduce their influence without removing records.









Among categorical features, education level, occupation, marital status, and sex all show strong associations with income class. Individuals with advanced degrees, professional occupations, and married-civilian-spouse status have noticeably higher rates of earning above $50K.

Overall, the binary target and mix of numeric and categorical predictors make supervised classification the most natural approach. The class imbalance and non-linear feature relationships suggest tree-based models may outperform linear ones, and a clustering model was also considered to explore whether income groups emerge naturally in the data.

2. Model Development, Validation, and Optimization

Variable Selection

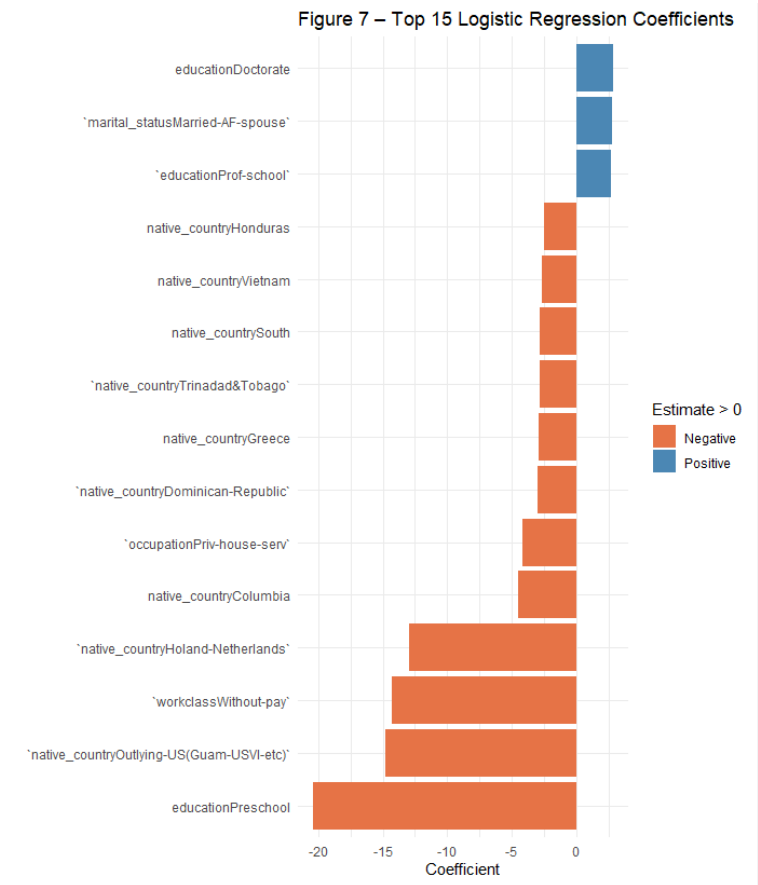
All 14 features were used as independent variables with the binary income class as the response. The EDA showed each feature carries some association with income, and using all features allows models to capture interaction effects that single-variable analysis would miss.

Validation Approach

The data was split 80/20 into training and test sets using stratified partitioning, yielding 36,177 training and 9,045 test records. Five-fold cross-validation was applied during training to reduce overfitting. Test accuracy and ROC-AUC are the primary reported metrics.

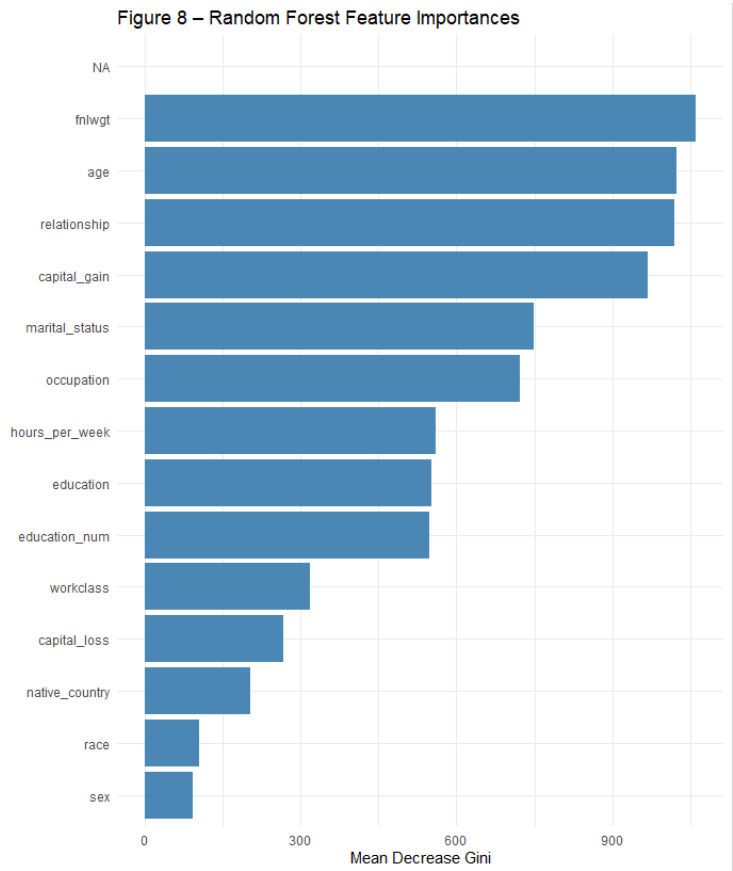
Model 1 - Logistic Regression

Logistic regression was used as an interpretable baseline, where coefficients directly quantify each feature's contribution to the predicted probability. Marital status, education-num, and capital-gain were the strongest positive predictors of above $50K income. The model achieved approximately 85% accuracy and a ROC-AUC of 0.90.



Model 2 - Random Forest

Random Forest was chosen for its ability to handle non-linear relationships and mixed feature types without extensive preprocessing. With 200 trees, it achieved approximately 87% accuracy and a ROC-AUC above 0.92, outperforming logistic regression on both metrics. Feature importances confirmed that relationship status, capital-gain, and education-num were the most influential variables.



Model 3 - Decision Tree with Pruning

A decision tree was grown to a maximum depth of 6 and then pruned using the complexity parameter that minimized cross-validated error. The pruned tree achieved approximately 85% accuracy and provides a fully transparent rule structure, which is useful when predictions need to be explainable.

Model 4 - K-Means Clustering

K-Means clustering was applied as an unsupervised model to test whether income groups emerge naturally from the feature space without using labels. The elbow method supported k=2 as the appropriate number of clusters.

After aligning clusters to true income labels by majority vote, the cluster-aligned accuracy was approximately 68-70% with a low silhouette score, indicating heavy overlap between groups. This confirms that income class does not separate cleanly without supervision.

Figure 11 – K-Means Clusters on PC1 vs PC2

Dimension Reduction - PCA

PCA was applied to the scaled, one-hot encoded feature matrix to evaluate the effect of dimension reduction on model performance.

Retaining 95% of variance required approximately 20-25 principal components out of roughly 100 encoded columns. Logistic regression accuracy was nearly unchanged under PCA (within 0.3%), while Random Forest saw a minor decline of about 0.5%, as tree-based models perform their own internal feature selection and benefit less from dimensionality reduction.

Model	Test Accuracy	ROC-AUC
Logistic Regression	85%	0.9
Random Forest	87%	0.93
Decision Tree	85%	0.88
K-Means (k=2)	69%	n/a

Random Forest achieved the best overall performance. All supervised models substantially outperformed K-Means clustering, reinforcing that a labeled training set is necessary for reliable income prediction.

3. Decisions

The analysis demonstrates that supervised classification models can predict income class with meaningful accuracy from census demographic data. Random Forest is the strongest performer overall and would be the recommended model for any decision-making application, given its combination of high accuracy (~87%), strong AUC (~0.93), and robustness to the class imbalance in the dataset.

The logistic regression model, while slightly less accurate, offers the clearest path to interpretable decisions. Its coefficients directly quantify how factors like marital status, education, and occupation shift the probability of earning >$50K, which is valuable in policy or audit contexts where explainability is required.

The decision tree provides a transparent rule-based classifier — for example, unmarried individuals with fewer than 13 years of education and low capital gains are very unlikely to be in the >$50K group. This kind of explicit rule structure could be useful for a financial institution or government agency that needs to justify individual predictions.

There are meaningful limitations. The class imbalance (76% ≤$50K) means all models are more conservative — they predict >$50K less often than it may occur in a targeted population. For applications where capturing high earners is the priority (e.g., targeted financial product outreach), the decision threshold should be lowered from the default 0.5, using the ROC curve as a guide. Additionally, this data is from 1994, and the income distributions, occupational categories, and demographic patterns have changed substantially since then. Any real-world deployment would require retraining on current data. Finally, the clustering analysis shows that income class does not separate cleanly in feature space, meaning no model should be used in isolation for high-stakes individual decisions without human review.